



AIAA 95-2970

**Numerical Analysis of Existing Liquid Rocket
Engines as a Design Process Starter**

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**31st AIAA/ASME/SAE/ASEE
Joint Propulsion Conference and Exhibit
July 10-12, 1995/San Diego, CA**

Numerical Analysis of Existing Liquid Rocket Engines as a Design Process Starter

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This paper uses multiple regression techniques to fit equations to historical and current liquid rocket engine designs. General equations are developed to estimate physical engine dimensions, dry mass, fuel and oxidizer mass flow rates, and turbopump pressure and power requirements. The resulting equations allow a simple approximation of a new engine design, with a desired thrust, and a desired fuel and oxidizer type, as the only input required. Based on modeling the existing engines with the design equations, the resulting approximation contains an inaccuracy of ± 10 to 20 percent. A discussion of the data fitting techniques, as well as tables of 30 design parameters for 14 existing liquid rocket engine designs are included.

BACKGROUND

Though they have been in use for over fifty years, the design and manufacture of liquid rocket engines is still time consuming and expensive. Reference 1 lists the amount of time from the beginning of the development phase to the operational phase for the Space Shuttle Main Engine (SSME) as 9 years, the Ariane Vulcain engine (HM60) as 8 years, and Pratt and Whitney's RL-10 as 5 years. The cost and complexity of liquid rocket engine development programs was evident in the planned development of the Space Transportation Main Engine (STME), which involved the teaming of three of America's largest aerospace firms, GenCorp's Aerojet Propulsion Division, Rockwell International's Rocketdyne Division, and United Technologies' Pratt and Whitney Government Engines and Space Propulsion Unit to design a new generation engine that would increase reliability and decrease cost.² Yet even with the combined resources of these companies, the STME program is virtually at a stand still.^{3,4}

INTRODUCTION

The purpose of this analysis is to provide simple, back-of-the-envelope equations to estimate liquid rocket engine general design parameters. This process is not a new one, Huzel and Huang did a much more complete and analytical analysis in their work from the 1960s.⁶

The scope of this analysis is limited to bi-propellant, turbopump fed, engines from approximately 60.0 KN to 8000.0 KN thrust. Fuel types of liquid hydrogen (LH_2), petroleum based (RP-1), and hydrazine blends were considered, as well as the oxidizers liquid oxygen (LO_2) and nitrogen tetroxide (N_2O_4). The paper assumes that the reader has decided on the fuel and oxidizer to be used and knows the desired thrust for the new engine. Additional design parameters are solved for to create a set of general parameters that describe the new engine.

ENGINE DATA

The following section lists the engines used in this analysis. A brief description of each of the engines follows.

HM7B: Built by Societe Europeenne de Propulsion (SEP) the engine is based on the first cryogenic engine developed in Europe, the HM7. Designed for the Ariane 3rd stage, development started in 1973 on the HM7A and qualification was in 1978. Using LH_2 fuel and LO_2 oxidizer and a gas generator cycle, the HM7B operates at a chamber pressure of 3.6 MPa and produces 62.2 KN of thrust. It uses a single turbine turbopump and a partially dump cooled nozzle. An upper stage engine, the HM7B is designed for operation at altitude.^{1,7,8,9,10,11}

RL10: Built by Pratt and Whitney, design started in 1958. The RL10 is the only expander cycle in this data set. It uses LH_2 and LO_2 and operates at a chamber pressure of 3.2 MPa. Though listed as an "old" engine in the tables because of its initial design date, the RL10 has been through many design iterations. The most recent, the RL10A-4, produces 92.5KN vacuum thrust and has a mass of 261 kg. Further design improvements could boost thrust to near 154KN in the future. The RL10 has been used in many applications, including the upper stage engine for the Saturn I and the Centaur. More recently it has been used as the engine on the Delta Clipper (DC-X) SSTO demonstrator. The RL10 version used in this analysis is the RL10A-3-3A. The RL10A-3-3A produces 73.4 KN of thrust and uses a single turbopump. The RL10A-3-3A is designed to operate at altitude.^{1,12-17}

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LE5: Built by Mitsubishi, LE5 development started in 1977. The LE5 was designed as the second stage engine for the H-1 booster. It uses LH_2 as the fuel and LO_2 as the oxidizer. It operates at a chamber pressure of 3.65 MPa and produces 103 KN of thrust. The LE5 uses a gas generator cycle with a single gas generator driving separate fuel and oxidizer turbopumps. The oxidizer turbine is downstream of the fuel turbine. A regeneratively cooled chamber and nozzle is used. The LE5 is designed to operate at altitude.^{1,18-20}

S-4: Built by Rocketdyne as the sustainer of the three engine Atlas first stage engine package, the MA-3, development dates back to 1954. The S-4 uses RP-1 as the fuel and LOX as the oxidizer, and produces 364 KN of thrust at a chamber pressure of 4.6 MPa. A gas generator cycle is used with one turbopump. A regeneratively cooled chamber and nozzle is used. The S-4 is designed to operate from sea level to altitude.^{11,21-26}

LR91: Built by Aerojet, development started approximately 1961. The LR91 is used on the Titan second stage and uses the storable propellants N_2O_4 and Aerozine-50, a 50/50 mix of hydrazine and unsymmetrical dimethylhydrazine. The LR91 operates at a chamber pressure of 5.9 MPa and produces 467 KN of thrust. A gas generator cycle is used with a single turbopump. The thrust chamber and a portion of the nozzle are regeneratively cooled, while the remainder of the nozzle (expansion ratio 14 to 49) uses an ablative skirt. The LR91 is designed to operate at altitude.^{27,28}

5C: Built by SEP, development started approximately 1973. The 5C is used on the Ariane 4 first stage and uses the storable propellants N_2O_4 and UH25, a 75/25 mix of unsymmetrical dimethyl hydrazine and hydrazine hydrate. The 5C operates at a chamber pressure of 5.8 MPa and produces 758 KN of thrust. A gas generator cycle is used with a single turbopump. The thrust chamber and nozzle are cooled using a thin film of fuel rather than the more typical regenerative cooling. The 5C is designed to operate from sea level to altitude.^{7,8}

H-1: Built by Rocketdyne for the Saturn I first stage, the H-1 was closely related to the Thor missile engine. Development began in 1958. It operates at a chamber pressure of 4.12 MPa and produces 945 KN of thrust. The H-1 uses RP-1 and LO_2 and uses a gas generator cycle to drive a single turbopump. Because of the close parallels in thrust, mixture ratio, etc., and lack of any

better data, the turbopump described in Ref. 6 was assumed to be the H-1 turbopump. A regeneratively cooled chamber and nozzle is used.^{6,13,22,23}

HM60: Built by SEP of France the HM60 development started in 1984. Designed as the first stage engine for the Ariane 5, the HM60 uses LH_2 and LO_2 . It operates at a chamber pressure of 10 MPa and produces 1025 KN of thrust. A gas generator cycle is used with a single gas generator supplying turbine power to separate fuel and oxidizer turbopumps. Like the HM7B, the HM60 uses a partially dump cooled nozzle. The HM60 is designed to operate from sea level to altitude.^{1,7,8,29}

J-2: Built by Rocketdyne for the Saturn IB second stage, development started in 1960. The J-2 uses LH_2 and LO_2 to produce 1023 KN of thrust at a chamber pressure of 5.4 MPa. A gas generator cycle is used with the single gas generator driving separate fuel and oxidizer turbopumps. A regeneratively cooled chamber and nozzle is used. The J-2 was designed to operate at altitude.^{1,11,13,21-23}

RS-27: Built by Rocketdyne for the Delta booster first stage, the RS-27 is an outgrowth of the Thor missile and Saturn I (H-1) engines. Although it has been updated, the RS-27 is essentially a 1955 design. The RS-27 uses RP-1 and LO_2 and operates at a chamber pressure of 4.87 MPa to produce 1043 KN of thrust. A gas generator cycle is used with the gas generator driving a single turbopump. A regeneratively cooled chamber and nozzle is used. The RS-27 is designed to operate from sea level to altitude.^{11,23,30}

LE-7: Built by Mitsubishi as the first stage engine for Japan's H-II booster, development started approximately 1984. The LE-7 uses LH_2 as fuel and LO_2 as the oxidizer. It operates at a chamber pressure of 12.7 MPa and produces 1080 KN of thrust. A staged combustion cycle is used with separate fuel and oxidizer turbopumps. A regeneratively cooled chamber and nozzle is used. The LE-7 is designed to operate from sea level to altitude.^{11,18,22,31,32-35}

LR87: Built by Aerojet as the Titan first stage, development coincided with the LR91 development in 1961. The LR87 is similar in design to the smaller LR91, including storable fuel and oxidizer types, engine cycle, and partial ablatively cooled nozzle. The LR87 produces 1218 KN of thrust at a chamber pressure of

5.91 MPa. The LR87 is designed to operate from ignition at sea level to altitude.^{27,28,36}

SSME: Built by Rocketdyne for the Space Shuttle, development started in approximately 1972. The SSME uses LH_2 and LO_2 and operates at a chamber pressure of 20.5 MPa to produce 2090 kN of thrust. The engine is designed to support variable thrust from approximately 65 - 104 percent, 100% thrust was used for the tables. A staged combustion cycle is used with low and high pressure turbopumps for both fuel and oxidizer. The efficiency data in the tables is for the high pressure pumps and turbines, power requirements are for the high pressure pump turbines. A regeneratively cooled chamber and nozzle is used. The SSME is designed to operate from sea level to altitude.^{1,21,22,37,38}

F-1: Built by Rocketdyne for the Saturn V first stage, development began in 1959. Related earlier work on 1 million pound thrust engines began in 1955. Design is similar to H-1 and RS-27, a gas generator cycle with one gas generator driven turbine driving both fuel and oxidizer turbopumps, but on a much larger scale. The F-1 uses RP-1 and LOX and produces 7775.5 kN at a chamber pressure of 7.76 MPa. A regeneratively cooled chamber and nozzle is used. The portion of the nozzle between area ratio 10 and 16 is cooled by turbine exhaust gases. The F-1 was designed to operate from sea level to altitude.^{13,21-23,39,40}

The following tables show the various design parameters collected for each engine. Not all of the items shown are used to determine the design equations, but are included for reference. A dash (-) in the table indicates data was not available for the item.

Table 1 shows general engine performance data including thrust, I_{sp} , combustion chamber pressure (P_c), and area and mixture ratios. Thrust and I_{sp} are given for vacuum operation.

Table 2 gives overall length, exit diameter and maximum diameter, dry mass, and the design age for each engine. The design age was used as an indicator variable to show age of design, with 1970 taken as the cutoff between old and new designs.

Table 3 shows combustion chamber design specifications where available. The characteristic length (L^*) is defined as the volume of the combustion chamber divided by the area at the nozzle throat.

Contraction ratio is defined as the area of the main diameter of combustion chamber divided by the area at the throat.²⁰

Table 4 lists fuel and oxidizer injector characteristics for each engine. Injector pressure drops were listed where available. The feed system pressure drop was calculated as the total pressure difference between the pump outlet pressure and the chamber pressure. Fuel and oxidizer combustion chamber mass flow rates were listed where available.

Table 5 shows feed system pump parameters. Total pump flow rates are listed as well as discharge pressure and pump efficiencies. The number of pump stages is listed, with the inducer not counted as a stage. For the staged combustion engines, LE-7 and SSME, the main pump output pressure was used to calculate the feed system pressure drop in Table 4. The J-2 fuel turbopump is the only axial pump listed, all other pumps were centrifugal.

Table 6 lists feed system turbine parameters, including number of stages, pressure ratio, turbine efficiency, and power requirements. The type of stage is also listed where available, where VCI = Velocity Compounded Impulse, IMP = Impulse, PCI = Pressure Compounded Impulse, and RI = Reaction Impulse.

Table 7 lists gas generator parameters where applicable, N/A indicates the engine does not use the gas generator cycle. Note that because of a lack of data, the gas generator pressure for the S-4 is the injector end pressure.

ENGINE	THRUST (KN)	Isp (sec)	P _c (MPa)	AREA RATIO	MIX RATIO
HM7B	62.2	445.5	3.6	82.9	4.565
RL10A-3-3A	73.4	446.4	3.2	61.1	5.0
LE-5	103.0	450.0	3.65	140.0	5.5
S-4(MA-3)	364.0	308.7	4.6	25.0	2.27
LR91	467.0	316.0	5.9	49.2	1.85
5C	758.0	278.0	5.85	10.5	1.7
H-1	945.4	292.0	4.12	8.0	2.26
HM60	1025.0	433.5	10.0	45.0	5.1
J-2	1023.0	425.0	5.4	27.5	5.5
RS-27	1043.0	301.8	4.87	12.0	2.245
LE-7	1080.0	445.6	12.7	52.0	6.0
LR87	1218.0	301.0	5.91	15.0	1.91
SSME	2090.0	455.2	20.5	77.5	6.0
F-1	7775.5	304.8	7.76	16.0	2.27

Table 1 General Performance Parameters

ENGINE	LENGTH (m)	EXIT DIA / MAX DIA (m)	DRY MASS (KG)	DESIGN
HM7B	2.01	0.992 / 0.992	158.0	NEW
RL10A-3-3A	1.78	1.0 / 1.0	138.0	OLD
LE-5	2.7	1.65 / 1.65	255.0	NEW
S-4(MA-3)	2.41	1.22 / 1.22	470.4	OLD
LR91	2.81	1.63 / 1.63	582.0	OLD
5C	2.87	1.0 / 1.0	826.0	TBD
H-1	2.67	1.24 / 1.24	878.2	OLD
HM60	3.1	1.76 / 2.5	1719.0	NEW
J-2	3.38	2.05 / 2.05	1542.0	OLD
RS-27	3.77	1.44 / 1.69	1146.6	OLD
LE-7	3.2	- / 2.57	1714.0	NEW
LR87	3.84	1.5 / -	1120.8	OLD
SSME	4.24	2.39 / 2.39	3150.0	NEW
F-1	6.1	3.66 / 3.66	8436.8	OLD

Table 2 Physical Parameters

ENGINE	LENGTH (m)	DIA (m)	L* (m)	MASS (kg)	CONTRACTION RATIO
HM7B	0.283	0.180	0.68	69.0	2.78
RL10A-3-3A	0.335	0.262	0.95	-	2.95
LE-5	0.351	0.240	0.84	40.8	3.11
S-4(MA-3)	2.18	0.303	1.09	166.0	1.66
LR91	0.391	0.367	0.97	71.2	2.51
5C	-	0.5	1.29	478.0	1.3
H-1	2.18	0.53	0.983	331.0	1.67
HM60	0.426	0.415	0.84	430.0	2.99
J-2	0.4572	0.47	0.62	446.8	1.58
RS-27	2.34	0.52	0.99	415.0	1.62
LE-7	0.37	0.40	0.78	145.5	2.75
LR87	0.51	0.55	1.07	155.0	2.08
SSME	0.356	0.45	0.80	390.0	2.96
F-1	3.35	1.02	1.22	-	-

Table 3 Combustion Chamber Parameters

ENGINE	INJ P DROP (KPa)		FEED P DROP (KPa)		MASS FLOW (kg/s)		
	FUEL	OXI	FUEL	OXI	FUEL	OXIDIZER	TOTAL
HM7B	920	1110	1930	1400	2.26	11.64	13.9
RL10A-3-3A	558	317	3951	1076	2.8	14.0	16.8
LE-5	549	1215	2390	1540	3.24	19.5	22.74
S-4(MA-3)	490	745	2489	2241	33.8	83.1	116.9
LR91	1338	1303	3210	2770	50.8	95.2	146.0
5C	-	-	1250	1550	101.0	171.0	272.0
H-1	965	1076	2980	2180	96.5	232.9	329.4
HM60	-	-	5800	3000	34.0	198.0	232.0
J-2	683	1248	3220	2240	36.1	206.2	242.3
RS-27	427	793	2220	2380	102.4	244.4	346.8
LE-7	1038	4598	14300	4700	35.2	211.1	246.3
LR87	2000	1276	3980	2340	135.2	270.3	405.5
SSME	1200	3716	21600	6700	67.0	402	469.0
F-1	641	2100	5240	3240	742	1784	2526

Table 4 Injector Parameters

ENGINE	FLOW RATES (kg/s)			DISCHG PRESS (MPa)		STAGES		EFFICIENCY	
	FUEL	OXI	TOT	FUEL	OXI	FUEL	OXI	FUEL	OXI
HM7B	2.57	11.7	14.27	5.55	5.02	1	1	60.0	73.0
RL10A-3-3A	2.8	14.0	16.8	7.18	4.31	2	1	57.0/55.5	64.0
LE-5	3.59	19.7	23.29	6.04	5.19	1	1	61.2	65.3
S-4(MA-3)	36.9	84.1	121.0	7.05	6.80	-	-	-	-
LR91	53.5	96.3	149.8	9.11	8.67	1	1	60.0	66.5
5C	103.0	173.0	276.0	7.1	7.4	1	1	65.0	66.0
H-1	102.4	234.9	337.3	7.1	6.3	1	1	71	75
HM60	39.7	202.5	242.2	15.8	13.0	2	1	73.0	76.0
J-2	38.2	212.1	250.3	8.62	7.64	7	1	73.0	80.0
RS-27	111.3	251.8	363.1	7.09	7.25	1	1	71.8	77.9
LE-7	35.7	211.1	246.8	27.0	17.4	2	1/1	69.7	76.5/78.6
LR87	141.8	270.9	412.7	9.89	8.25	1	1	63.0	67.0
SSME	67.7	407.0	474.7	42.1	27.2	3	1/1	76.0	68.1/80.5
F-1	796	1804	2600	13.0	11.0	1	1	-	-

Table 5 Feed System Pump Parameters

ENGINE	STAGES - TYPE		PRESS RATIO		EFFICIENCY		POWER (KW)	
	FUEL	OXI	FUEL	OXI	FUEL	OXI	FUEL	OXI
HM7B	2-IMP	†	16.7	†	45.0	†	404	†
RL10A-3-3A	2-	†	1.41	†	72.0	†	563	†
LE-5	2-VCI	2-VCI	4.59	1.85	47.6	39.2	472	132
S-4(MA-3)	2-	†	-	†	-	†	1257	†
LR91	2-	†	21.2	†	53.0	†	1721	†
5C	2-	†	16.6	†	50.0	†	2495	†
H-1	2-PCI	†	18.21	†	66.0	†	2830	†
HM60	2-	1-	17.0	13.6	59.0	27.0	11200	3000
J-2	2-	2-	7.2	2.65	60.0	47.0	6403	1717
RS-27	2-PCI	†	22.0	†	58.9	†	3346	†
LE-7	1-IMP	1-IMP	1.49	1.38	68.9	47.9	18000	4500
LR87	2-	†	15.7	†	52.0	†	4200	†
SSME	3-RI	2-RI	1.44	1.55	84.2	76.9	55896	21058
F-1	2-IMP	†	16.3	†	60.5	†	40000	†

†Single turbine drives both pumps

Table 6 Feed System Turbine Parameters

ENGINE	FLOW RATE (kg/s)	COMB. TEMP (K)	COMB. PRESS. (MPa)	MIXTURE RATIO
HM7B	0.25	860.0	2.3	0.87
RL10A-3-3A	N/A	N/A	N/A	N/A
LE-5	0.436	837.0	2.63	0.85
S-4(MA-3)	4.00	843.8	5.15	0.297
LR91	2.71	1172	3.21	0.09
5C	6.36	888	3.66	2.4
H-1	7.86	922	4.22	0.342
HM60	8.4	871	8.5	0.9
J-2	3.19	922+	4.7	0.94
RS-27	9.13	916	4.7	0.33
LE-7	N/A	N/A	N/A	N/A
LR87	7.23	1255	4.3	0.084
SSME	N/A	N/A	N/A	N/A
F-1	75.7	1062	6.76	0.416

Table 7 Gas Generator Parameters

ANALYSIS

Once the engine data had been gathered, the analysis involved the use of multiple regression techniques to fit multivariable equations to the engine design parameters. From one up to six independent variables were used to model the dependent variables. Since the analysis assumed that only thrust and desired propellant type were available initially, subsequent design equations were limited to these parameters, and any predicted parameters, as independent variables. The general model for multiple linear regression is written as

$$y_i = B_0 + \sum_{j=1}^k B_j x_{ij} + \varepsilon_i \quad i = 1, 2, \dots, n \quad (1)$$

where y_i is the dependent variable, $B_0, B_1, \dots, B_n$ are the regression coefficients, x_{ij} is the independent variable, ε_i is random error, k is the number of independent variables, i is the measurement number and n is the number of measurements. Eq. (1) can be rewritten as

$$(\varepsilon)^2 = \left(y_i - B_0 - \sum_{j=1}^k B_j x_{ij} \right)^2 \quad (2)$$

and solved by taking the partial derivative with respect to $B_0, B_1, \dots, B_n$ and setting each equal to zero to find the

regression coefficients. Eq. (1) can be more conveniently expressed in matrix notation as

$$\mathbf{Y} = \mathbf{XB} + \mathbf{E} \quad (3)$$

where $\mathbf{Y}$ is the n by 1 column matrix of dependent variables, $\mathbf{E}$ is the n by 1 column matrix of random error, $\mathbf{B}$ is a $k+1$ by 1 column matrix of regression coefficients, and $\mathbf{X}$ is the n by $k+1$ matrix of independent variables. Eq. (2) can then be rewritten in matrix form as

$$\mathbf{E}^2 = \mathbf{Y}^T \mathbf{Y} + \mathbf{B}^T \mathbf{X}^T \mathbf{X} \mathbf{B} - 2\mathbf{B}^T \mathbf{X}^T \mathbf{Y} \quad (4)$$

Taking the derivative of Eq. (4) with respect to $\mathbf{B}$ and setting the result equal to zero yields the solution

$$\mathbf{B} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{Y} \quad (5)$$

where $\mathbf{B}$ is the $k+1$ by 1 matrix of regression coefficients.⁴³ Because the regression models assume a linear relationship between the independent and dependent variables, transformations of the independent and dependent variables using log and natural log conversions were employed to linearize the data where required.^{41,42} Polynomial approximations of the dependent variable were not considered because of the

ability of an $n-1$ order polynomial to pass through n points, yet not yield any real prediction value.⁴³ The independent variables included major design parameters such as thrust, chamber pressure, mass flow rates, etc. The dependent variables included engine mass, overall engine dimensions, propellant flow rates, feed system pressure drop, and feed system power requirements. Indicator variables, discrete regression coefficients that are set to either 1 or 0 and allow a difference in slope between two sets of data, were used to identify engine ignition altitude, high/low I_{sp} , and age of design.⁴¹⁻⁴³ Other than its use as an indicator variable, I_{sp} was not used as an independent variable because it was considered a detailed design parameter, rather than the more general design parameters that are the focus of this analysis. Note that the high/low I_{sp} indicator, with a threshold of 400 sec., is essentially a LH_2 fuel type indicator.

Because of the large magnitude difference in the variables used, for example thrust (60-8000KN) versus area ratio (10-150), the principle of standardized variables was incorporated. Unit normal scaling was selected where a unit normal standardized variable z_{ij} , is given by

$$z_{ij} = (x_{ij} - x_j) / s_j \quad (6)$$

where x_{ij} is the known variable, x_j is the average for the variable type, and s_j is the standard deviation for the variable type. Unit normal standardization was applied to both the dependent and independent variables.^{43,44}

Once an initial model was generated using obvious first choices, for instance the use of thrust when fitting engine mass, an iterative procedure was used to select independent variables that decreased the amount of error between the predicted and actual measurement. The criteria used to determine an improvement in the fit included a decrease in the residual mean square (RMS) of the error, a significant F-value, and an increase in the multiple correlation coefficient. The multiple correlation coefficient (R^2) is defined as the amount of variation in the dependent variable that is predicted by the independent variable(s) or

$$R^2 = 1 - \left[\frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2} \right] \quad (7)$$

where y_i is the measurement, $\hat{y}_i$ is the predicted measurement, and $\bar{y}$ is the average over all the measurements.

While R^2 is a good indicator of the quality of the fit, it is a poor choice to determine which independent variables are significant because it usually increases each time an independent variable is added.⁴³ For this reason the F-value was monitored for each independent variable. The F-value calculates the percent error between the sum of residuals squared in the reduced model and the sum of residuals squared in the full model or

$$F = \frac{(SSRM - SSFM) / (p + 1 - k)}{SSFM / (n - p - k)} \quad (8)$$

where

$$SSFM = \sum (y_i - \hat{y}_i)^2 \quad (9)$$

and

$$SSRM = \sum (y_i - \hat{y}'_i)^2 \quad (10)$$

In Eq. (8) p is defined as the number of variables in the full model, n is defined as the number of measurements, and k is the number of variables in the reduced model. In Eq. (9) $\hat{y}_i$ is the predicted measurement, and in Eq. (10) $\hat{y}'_i$ is the measurement predicted without using the variable for which the F-value is being calculated. A variable is assumed to be significant (passes the F-test) when the F-value is greater than the value in the F distribution at k , and $n-k-1$, degrees of freedom.^{42,44} In a few cases, where only a limited number of measurements were available because of the separation of data sets, the F-value of an independent variable would not pass the F-test. In these cases the RMS value was also used to determine whether or not a variable was improving the fit. A PC-based software implementation of Eq. (5) was used to fit the design equations to the data.

ENGINE DESIGN EQUATIONS

The first design parameter considered was combustion chamber pressure. Because desired thrust is the only parameter assumed to be known at this point in the analysis, the fit was specifically limited to thrust as the only independent variable. A natural logarithm transformation was used on both chamber pressure (P_c)

and thrust. In this case the old and new engines had to be divided up into separate data sets. An indicator variable was not enough to differentiate between the old and new designs. The equations for chamber pressure (in MPa) as a function of thrust (KN) are

for a modern design

$$P_c = 0.4315 * (\text{THRUST})^{0.48284} \quad (11)$$

for an older design

$$P_c = 1.6330 * (\text{THRUST})^{0.17283} \quad (12)$$

The next step was to look at parameters important to the turbomachinery design process. Propellant mass flow rates, pressure difference between pump discharge and chamber pressure, and turbine power requirements were assumed to provide a gross definition of the turbopump design. Total pressure drop between the pump discharge and the combustion chamber was used so the pump discharge pressure could be estimated from just the chamber pressure. Total engine mass flow rate ($\dot{m}_{tot}$) was the first parameter considered. Total engine mass flow rate was fit using thrust and an indicator variable for $I_{sp} > 400$ sec. as the independent variables.[‡] A natural logarithm transformation was used on both thrust and $\dot{m}_{tot}$. The design equation for $\dot{m}_{tot}$ (in kg/sec) as a function of thrust (KN) and the I_{sp} indicator is

$$\dot{m}_{tot} = 0.3290 * (\text{THRUST})^{1.00542} * I \quad (13)$$

where $I = 1$ if $I_{sp} < 400$ and $I = 0.6815$ if $I_{sp} > 400$.

The equation for fuel mass flow rate ($\dot{m}_{fuel}$) was determined next. Thrust and an indicator variable to specify the fuel type were used as the independent variables. A natural logarithm transformation was used on both thrust and $\dot{m}_{fuel}$. The design equation for $\dot{m}_{fuel}$ (in kg/sec) as a function of thrust (KN), and the fuel type indicator is

$$\dot{m}_{fuel} = 0.1453 * (\text{THRUST})^{0.97352} * I \quad (14)$$

where $I = 1$ for hydrazine based fuels, $I = 0.8701$ for RP-1, and $I = 0.2933$ for LH₂.

[‡]Alternatively, $\dot{m}_{tot}$ could be estimated by guessing at the I_{sp} and dividing the thrust by $I_{sp} * g$.

The oxidizer mass flow rate ($\dot{m}_{ox}$) can be predicted by following the method used for $\dot{m}_{fuel}$. As in Eq. (14), a natural logarithm transformation was used on both thrust and $\dot{m}_{ox}$. The design equation for $\dot{m}_{ox}$ (in kg/sec) as a function of thrust (KN) and the fuel type indicator is

$$\dot{m}_{ox} = 0.2199 * (\text{THRUST})^{1.01130} * I \quad (15)$$

where $I = 1$ for RP-1, $I = 0.9237$ for hydrazine based fuels, and $I = 0.8301$ for LH₂.

The next step in the turbomachinery design process involved determining the feed system pressure drop between the pump outlet and the combustion chamber. Chamber pressure was found to be the most significant independent variable for the fuel feed system pressure drop. An indicator variable was also included to separate engines that do not use the gas generator cycle. Two engines were removed from the initial data set as outliers, the RL10 and the 5C. The RL10 was removed because it was the only expander cycle engine and therefore had a much greater fuel pressure drop. The 5C was removed because it had a much lower pressure drop than all the other engines, most probably due to its use of film cooling rather than regenerative cooling. The removed engines can be seen in Figure 1 marked with the $\times$ plotting symbol.

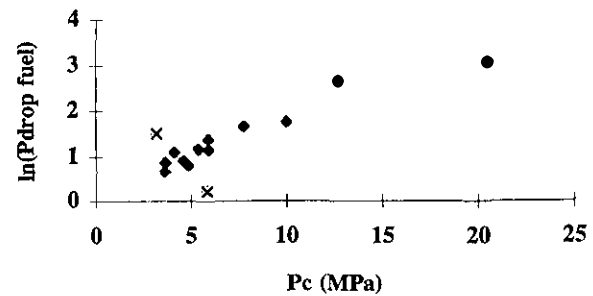


Figure 1 Fuel Pressure Drop

A natural logarithm transformation was used on both the independent and dependent variables. The design equation for fuel feed system pressure drop (in MPa) as a function of P_c (MPa) and a gas generator/staged combustion indicator variable is

$$P\text{-drop (fuel)} = 0.5829 * (P_c)^{1.00919} * I \quad (16)$$

where $I = 1$ for gas generator engines and $I = 1.821$ for staged combustion engines.

For the fit of oxidizer feed system pressure drop chamber pressure was again found to be the most significant independent variable. An indicator variable was included to separate the gas generator and staged combustion engines. The RL10 and 5C engines were removed from the data set. In this case both engines had low oxidizer pressure drops. A natural logarithm transformation was used on the dependent variable only.

The design equation for the oxidizer feed system pressure drop (in MPa) as a function of P_c (MPa) is

$$P\text{-drop (ox)} = -0.6440 + 1.7815 * \ln(P_c) + I \quad (17)$$

where $I = 1$ for gas generator engines, and $I = 1.1895$ for staged combustion engines.

Having determined the feed system pressure drops, the final step in the turbopump design process was to determine the turbine power required to drive the pumps. The required turbine power was assumed to be the sum of both the fuel and oxidizer pump power requirements. Gearbox losses (if any) were assumed to be negligible. Figure 2 shows the actual turbine power for the engines in this data set. Two distinct groups of data are evident, the large high I_{sp} (LH_2 fueled) engines marked with the ■ plotting symbol, and the remaining engines marked with the ◆ plotting symbol.

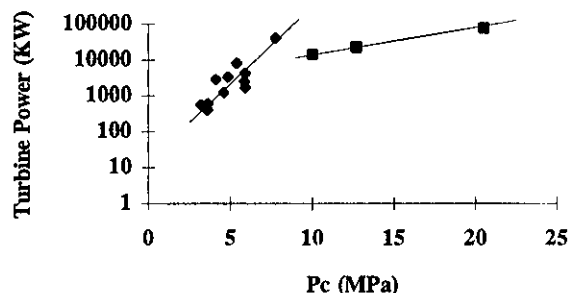


Figure 2 Actual Turbine Power

For turbine power, the most significant independent variables were chamber pressure, $\dot{m}_{tot}$ and an indicator variable for $I_{sp} > 400$ sec. A natural logarithm transformation was used on $\dot{m}_{tot}$ and the dependent variable. The design equation for total turbine power (in KW) versus P_c (MPa) and $\dot{m}_{tot}$ (kg/sec) is

$$Power = 5.6069 * (1.1120)^{P_c} * (\dot{m}_{tot})^{1.00452} * I \quad (18)$$

where $I = 1$ for $I_{sp} < 400$, and $I = 3.5114$ for $I_{sp} > 400$.

Eq. (18) shows that the high I_{sp} (LH_2 fueled) engines have turbine power requirements approximately 3.5 times the low I_{sp} engines.

The combustion chamber design parameters were evaluated next. Because of differences in the measurement of combustion chamber mass and length, only combustion chamber diameter was modeled.[§] The characteristic length (L^*) was initially considered as a design parameter to model. Further analysis showed that L^* had little correlation to the available independent variables. The equation for combustion chamber diameter, (CC_{dia}), was fit using P_c and $\dot{m}_{tot}$ as the independent variables. An indicator variable was used for $I_{sp} > 400$ sec. A natural logarithm transformation was used on both the independent and dependent variables. The design equation for CC_{dia} (in meters) as a function of P_c (MPa) and $\dot{m}_{tot}$ (kg/sec) is

$$CC_{dia} = 0.0927 * (P_c)^{-0.28329} * (\dot{m}_{tot})^{0.37214} * I \quad (19)$$

where $I = 1$ if $I_{sp} < 400$, and $I = 1.1427$ if $I_{sp} > 400$.

The physical engine design parameters were evaluated next. The equation for engine dry mass was determined using thrust and an indicator variable for $I_{sp} > 400$ sec. as the independent variables. Chamber pressure was found to be insignificant. A natural logarithm transformation was used on both the independent and dependent variables. The design equation for engine dry mass (in kg) as a function of thrust (KN), and the I_{sp} indicator is

$$Mass = 2.2890 * (THRUST)^{0.89255} * I \quad (20)$$

where $I = 1$ if $I_{sp} < 400$, and $I = 1.522$ if $I_{sp} > 400$.

The indicator variable in Eq. (20) shows that the high I_{sp} engines (LH_2 fuel), have a greater mass for a given thrust. This is also evident in Figure 3, which shows

[§] Depending upon the design, some combustion chamber lengths are defined as injector to throat, while others are defined as injector to nozzle exit. For mass, some combustion chambers are fabricated as a single unit including the nozzle. In these cases the combustion chamber mass is not available.

the thrust per unit mass versus thrust. Figure 3 shows that the LH₂ engines, marked with the ■ plotting symbol, have a thrust per unit mass ratio of ≈ 0.4 to 0.6 KN/kg while the RP-1 and hydrazine engines (◆) have a thrust per unit mass ratio of ≈ 0.7 to 1.0 KN/kg.

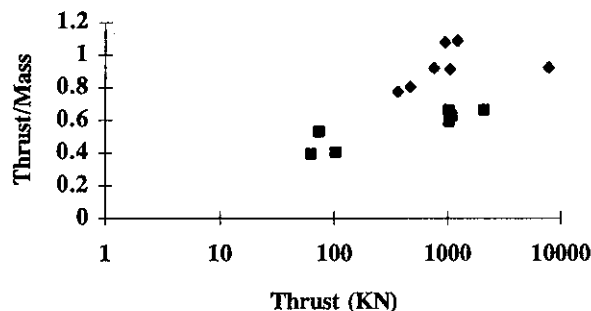


Figure 3 Thrust/Mass vs Thrust

The equation for overall engine length was determined using thrust (KN) and an indicator variable for sea level vs vacuum ignition altitude. A natural logarithm transformation of length and thrust gave the best results. The design equation for length (in meters) as a function of thrust (KN) and the ignition altitude indicator is

$$\text{Length} = 0.5936 * (\text{THRUST})^{0.24750} * I \quad (21)$$

where $I = 1$ if the design is for sea level ignition, and $I = 1.1405$ if the design is for ignition at altitude.

The indicator variable in Eq. (21) shows that the higher Area Ratio on the vacuum ignition engines increases the length by approximately 14 percent.

The equation for engine exit diameter (D_e) used thrust, and two indicator variables, an old/new design indicator variable and an ignition altitude indicator. A natural logarithm transformation was used on both exit diameter and thrust. The LE-7 engine was not included because exit diameter data was not available. The design equation for exit diameter (in meters) as a function of thrust (KN), area ratio and design age is

$$D_e = 0.1238 * (\text{THRUST})^{0.35247} * I_1 * I_2 \quad (22)$$

where $I_1 = 1$ if the engine is an old design, and $I_1 = 1.319$ if the engine is a new design. $I_2 = 1$ if the design is for sea level ignition, and $I_2 = 1.610$ if the design is for ignition at altitude.

The design equations for the physical parameters, (Eqs. 20-22) show a strong dependence on Area Ratio. Using Area Ratio as an independent variable decreased the error in the predictions for mass, length and diameter for all the engines. Unfortunately, Area Ratio showed little correlation with any of the independent variables and was predictable to within 20 percent on only 8 of the 14 engines. The remaining engines had errors in Area Ratio estimation of up to 50 percent. Reference 6 shows that theoretical Area Ratio can be calculated from

$$\frac{A_e}{A_t} = \frac{\left(\frac{2}{(\gamma + 1)} \right)^{\frac{1}{\gamma-1}} [P_c / P_e]^{\left(\frac{1}{\gamma} \right)}}{\sqrt{\frac{\gamma + 1}{\gamma - 1} \left[1 - (P_e / P_c)^{\frac{\gamma-1}{\gamma}} \right]}} \quad (23)$$

where A_e is the nozzle exit area, A_t is the throat area, P_e is the nozzle exit pressure, and γ is the ratio of specific heat for the combustion products. Eq. (23) could be used to estimate the Area Ratio for the new design. However, Area Ratio is often a tradeoff between various design considerations, including trajectory profile, nozzle mass, and thermal considerations, and is rarely taken directly from a theoretical calculation.⁶ Because of this inability to predict Area Ratio additional error is introduced into the equations for the physical engine parameters.

LIMITATIONS

The greatest limitation of this process is that regression analysis is only valid as an interpolation over the range of the independent variables. Extrapolation outside of the range can lead to large errors.⁴¹⁻⁴³ In this analysis, predictions in the 3000 KN to 8000 KN thrust range may also be questionable because of the single data point (the F-1) at ≈ 8000KN. The lack of a large number of data points is not solely the fault of this analysis; there simply are not many large liquid rocket engines in existence. For instance, no new design American RP-1/LO₂ engines were found in the research for this paper. The Commonwealth of Independent States (CIS) has a large number of engines in production, but due to a lack of points of contact for the CIS engines, this analysis was limited to American, European and Japanese engines. Using out of production engines helps, but does not completely remove this limitation.

Data quality is also a limitation. Several manufactures of engines, and their customers, responded to requests to provide data for this analysis, but the data for the H-1, J-2, and F-1 engines had to be determined from several different sources. Often the data between sources does not agree and it is impossible to determine which is more accurate.

CONCLUSION

The purpose of this analysis was to provide simple equations to estimate complex designs. The shortcoming of this is that the equations are averaging across all the engines used as independent variables. Using the equations to predict the designs from which they were generated shows that they are only valid to approximately ± 10 to 20 percent of the actual designs. Obviously this limits their use to very preliminary design work. An increase in the number of engines, and a validation of the data for some of the current engines, would reduce this inaccuracy. A method to estimate practical, rather than theoretical, Area Ratio is also required to improve the accuracy of the physical engine parameter equations.

ACKNOWLEDGMENTS

The author wishes to thank the following people; Mr. Hideaki Katsuta of the National Space Development Agency of Japan for invaluable data on the LE-5 and LE-7 engines, Mr. Marcel Pouliquen of SEP and Mr. Pascal Pempie of CNES for data on the HM7B, 5C and HM60 engines, Mr. Jim Brown of Pratt and Whitney for data on the RL10, Mr. Mark Fisher of Marshall Space Flight Center for data on the J-2, SSME, and F-1 engines, Mr. Larry Hansen and Gary Cook of Aerojet for data on the LR91 and LR87 engines, and Dr. Ron Humble for the original idea.

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